1 The diagram shows a triangle.

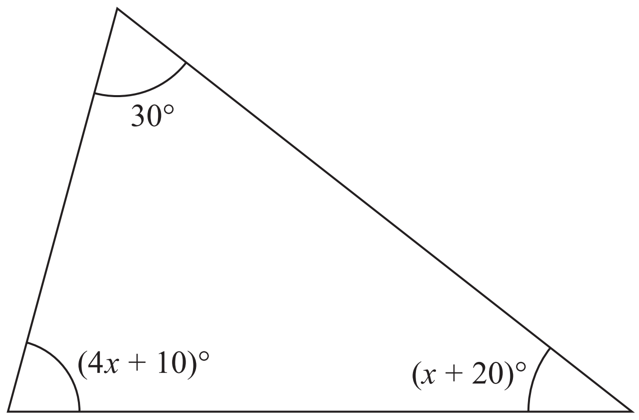


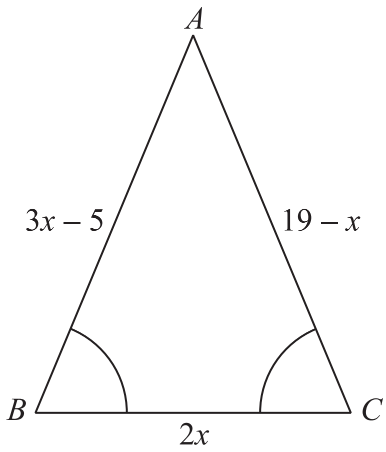
Diagram **NOT** accurately drawn

Work out the value of x.

$x =$ ....

(Total for Question 1 is 4 marks)

2 ABC is a triangle.



Angle ABC = angle BCA.

The length of side AB is $(3x - 5)$ cm.
The length of side AC is $(19 - x)$ cm.
The length of side BC is $2x$ cm.

Work out the perimeter of the triangle.
Give your answer as a number of centimetres.

.... cm

(Total for Question 2 is 5 marks)

3 The diagram shows the triangle PQR.

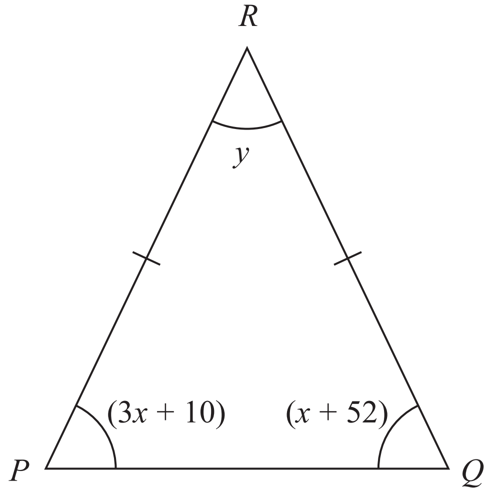


Diagram **NOT** accurately drawn

In the diagram, all the angles are in degrees.

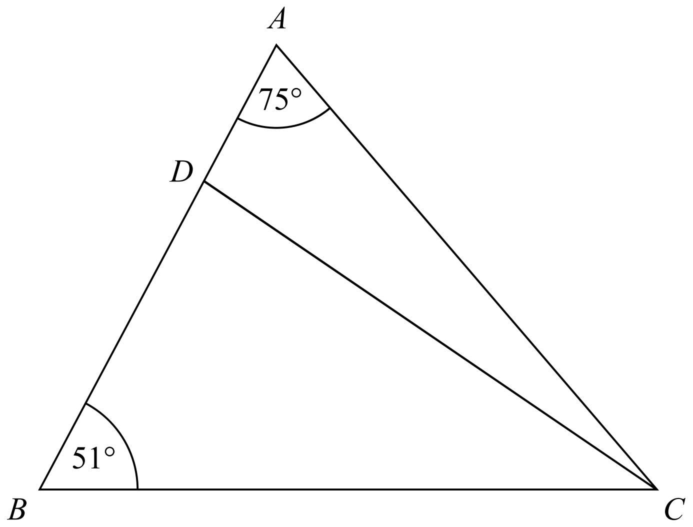
$RP = RQ$

Find the value of y.
Show clear algebraic working.

$y =$ ....

(Total for Question 3 is 4 marks)

4 The diagram shows triangle *ABC*.



ADB is a straight line.

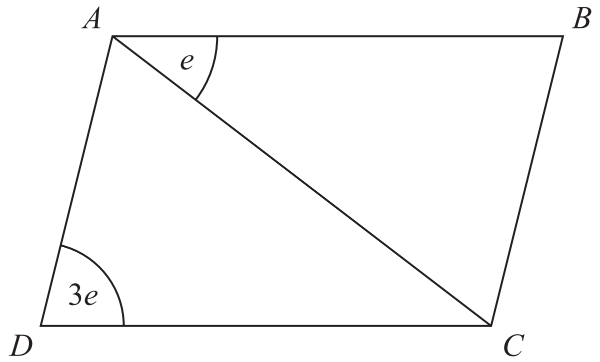
the size of angle *DCB* : the size of angle *ACD* = 2 : 1

Work out the size of angle *BDC*.

....°

(Total for Question 4 is 4 marks)

5 *ABCD* is a parallelogram.



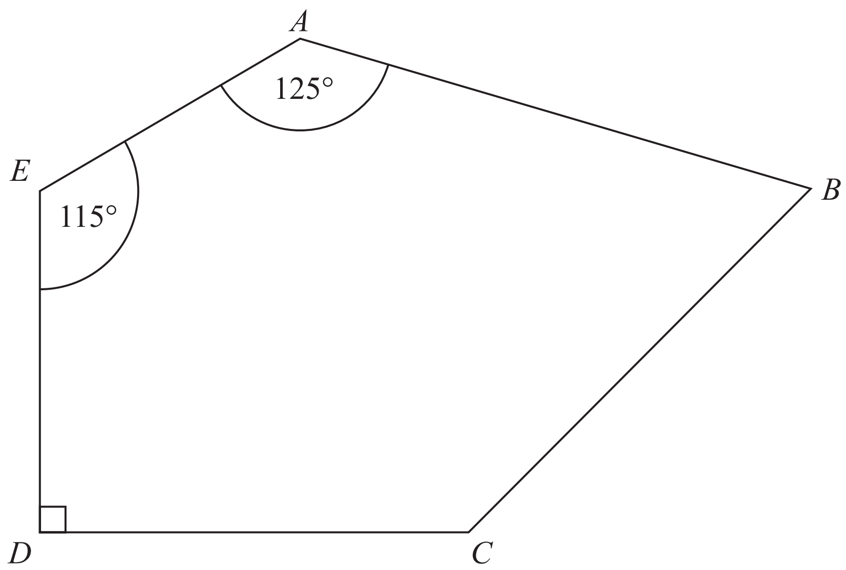
All angles are measured in degrees.

Find an expression, in terms of e, for the size of angle *CAD*.
Give a reason for each stage of your working.

....

(Total for Question 5 is 3 marks)

6 *ABCDE* is a pentagon.



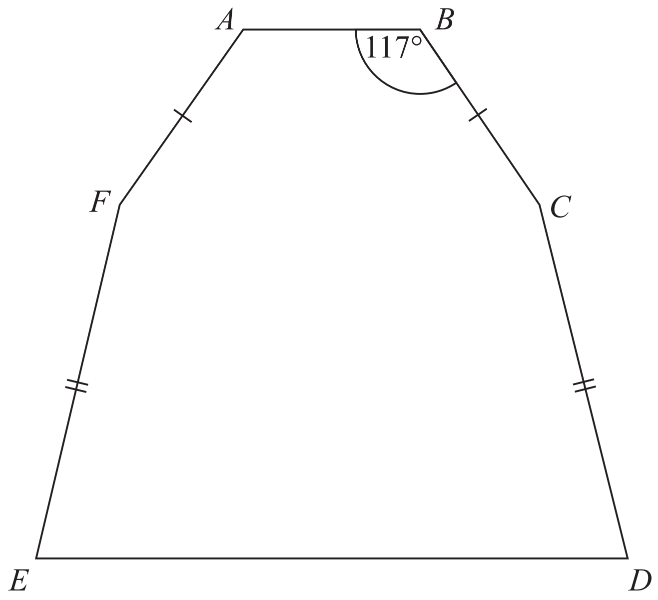
Angle *BCD* = 2 × angle *ABC*

Work out the size of angle *BCD*.
You must show all your working.

....°

(Total for Question 6 is 5 marks)

7 The diagram shows a hexagon.
The hexagon has one line of symmetry.



$FA = BC$
$EF = CD$
Angle $ABC = 117°$

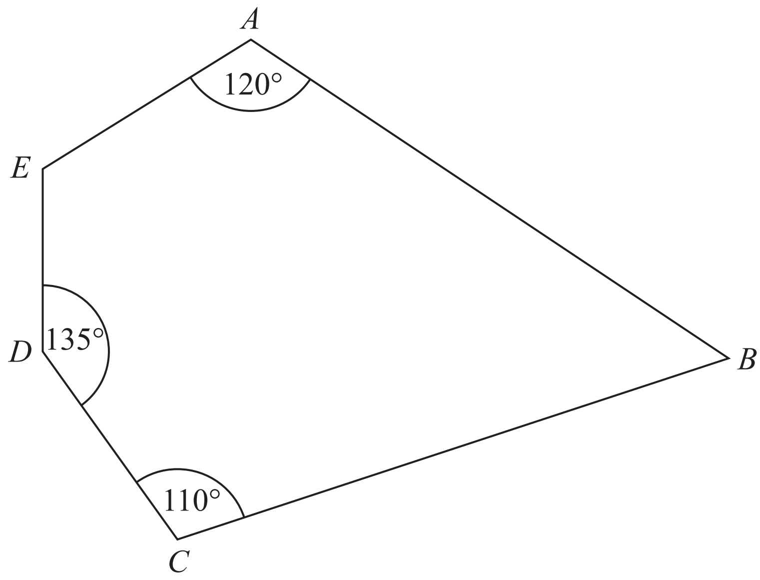
Angle $BCD = 2 \times$ angle CDE

Work out the size of angle AFE.
You must show all your working.

.... °

(Total for Question 7 is 4 marks)

8 Here is a pentagon.



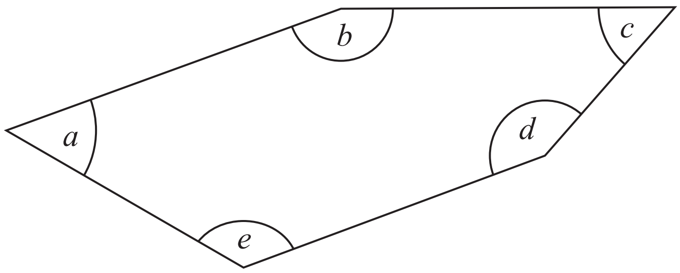
Angle $AED = 4 \times$ angle ABC

Work out the size of angle AED.
You must show all your working.

.... °

(Total for Question 8 is 4 marks)

9 Here is a pentagon.



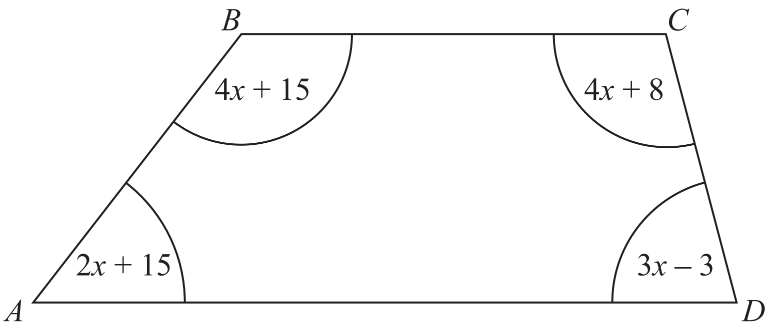
Angle a = angle c
Angle b = 155°
Angle d is three times the size of angle c
Angle e is two times the size of angle c

Work out the size of angle a

....°

(Total for Question 9 is 4 marks)

10 *ABCD* is a quadrilateral.



All angles are measured in degrees.

Show that *ABCD* is a trapezium.

(Total for Question 10 is 4 marks)

11 *ABCD* is a trapezium.

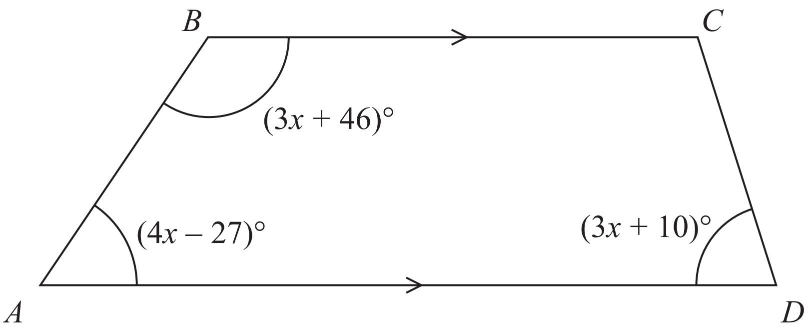


Diagram **NOT** accurately drawn

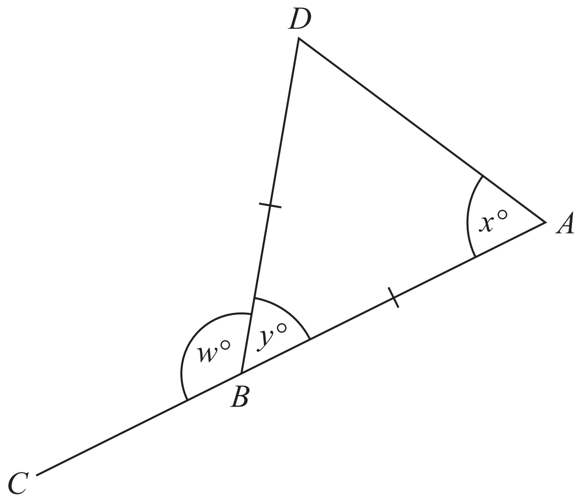
BC is parallel to *AD*

Find the size of the largest angle inside the trapezium.

.... °

(Total for Question 11 is 4 marks)

12 The diagram shows an isosceles triangle *ABD* and the straight line *ABC*.



$BA = BD$

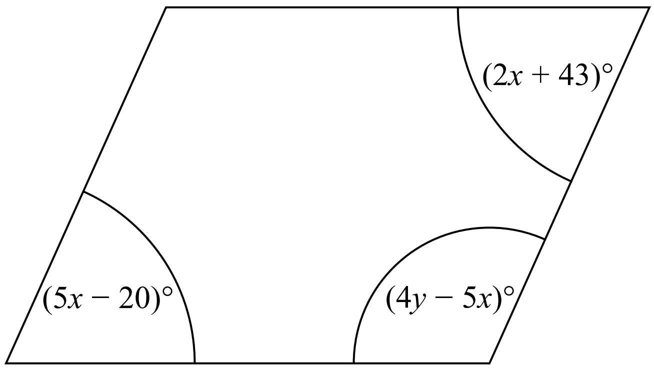
$x : y = 2 : 1$

Work out the value of w.

$w =$ ....

(Total for Question 12 is 4 marks)

13 Here is a parallelogram.



Work out the value of x and the value of y.

x = ...

y = ...

(Total for Question 13 is 5 marks)

14

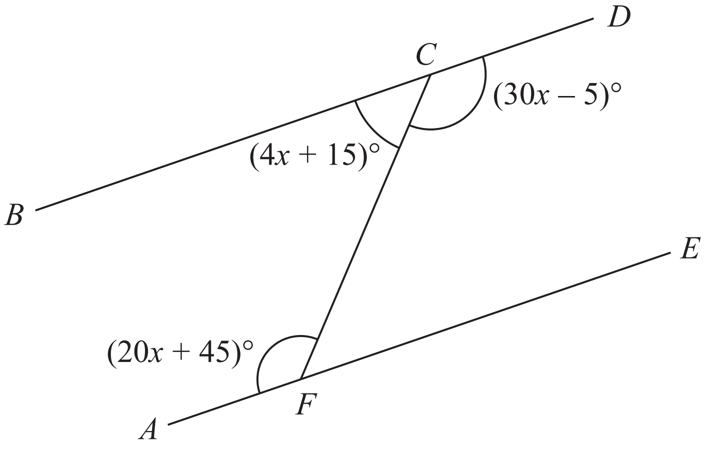


Diagram **NOT** accurately drawn

BCD and *AFE* are straight lines.

Show that *BCD* is parallel to *AFE*.
Give reasons for your working.

(Total for Question 14 is 5 marks)